Miscellaneous

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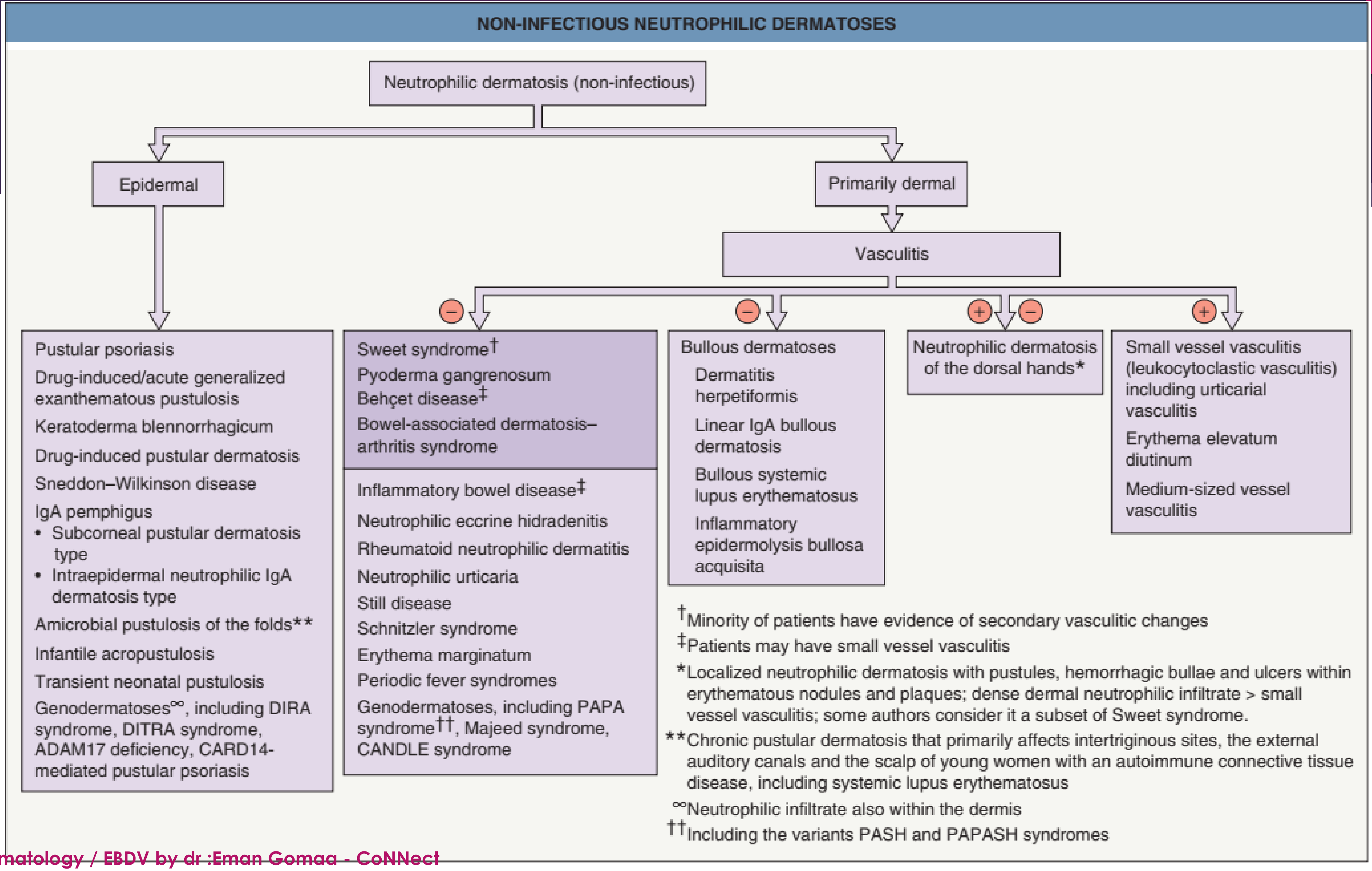
Neutrophilic dermatosis

- Dermal
  - With vasculitis
    - CSVV
    - HSP
    - UV
    - EED
    - PAN
    - CRUOGLOBULINAMIA
  - Without vasculitis
    - SWEET syndrome
    - PG
    - BULLOUS DERMATOSIS
    - DH
    - LABD
    - E,B acquisita
    - Bullous sle
    - behcet
- epidermal
  - Pustular ps
  - AGEP
  - Iga pemphigus


NON-INFECTIOUS NEUTROPHILIC DERMATOSES
Neutrophilic dermatosis (non-infectious)
Epidermal
Primarily dermal
Vasculitis
Pustular psoriasis
Drug-induced/acute generalized exanthematous pustulosis
Keratoderma blennorrhagicum
Drug-induced pustular dermatosis
Sneddon–Wilkinson disease
IgA pemphigus
• Subcorneal pustular dermatosis type
• Intraepidermal neutrophilic IgA dermatosis type
Amicrobial pustulosis of the folds**
Infantile acropustulosis
Transient neonatal pustulosis
Genodermatoses∞, including DIRA syndrome, DITRA syndrome, ADAM17 deficiency, CARD14-mediated pustular psoriasis
−
Sweet syndrome†
Pyoderma gangrenosum
Behçet disease‡
Bowel-associated dermatosis–arthritis syndrome
Inflammatory bowel disease‡
Neutrophilic eccrine hidradenitis
Rheumatoid neutrophilic dermatitis
Neutrophilic urticaria
Still disease
Schnitzler syndrome
Erythema marginatum
Periodic fever syndromes
Genodermatoses, including PAPA syndrome††, Majeed syndrome, CANDLE syndrome
−
Bullous dermatoses
Dermatitis herpetiformis
Linear IgA bullous dermatosis
Bullous systemic lupus erythematosus
Inflammatory epidermolysis bullosa acquisita
+ −
Neutrophilic dermatosis of the dorsal hands*
+
Small vessel vasculitis (leukocytoclastic vasculitis) including urticarial vasculitis
Erythema elevatum diutinum
Medium-sized vessel vasculitis
†Minority of patients have evidence of secondary vasculitic changes
‡Patients may have small vessel vasculitis
*Localized neutrophilic dermatosis with pustules, hemorrhagic bullae and ulcers within erythematous nodules and plaques; dense dermal neutrophilic infiltrate > small vessel vasculitis; some authors consider it a subset of Sweet syndrome.
**Chronic pustular dermatosis that primarily affects intertriginous sites, the external auditory canals and the scalp of young women with an autoimmune connective tissue disease, including systemic lupus erythematosus
∞Neutrophilic infiltrate also within the dermis
††Including the variants PASH and PAPASH syndromes

Sweet syndrome
acute febrile neutrophilic dermatosis

- Uncommon disease
- Mor in female 4:1
- Worldwide Japan

ogenesis

Genatic
C 3q
HLA54

→

HYPERSENSETIVITY REACTION

→

IMMUNE DYSREGULATION

SWEET SYNDROME – ASSOCIATED DISORDERS AND TRIGGERS
Infections
• Viruses: **upper respiratory tract infections**, cytomegalovirus, hepatitis B virus, hepatitis C virus, HIV • Bacteria: *Yersinia*, streptococci • Mycobacteria: atypical mycobacteria, BCG vaccination, *M. tuberculosis*, *M. leprae* • Fungi: dimorphic, including sporotrichosis and coccidiomycosis
Malignancies
• Hematologic (10–20% of cases*), in particular **acute myelogenous leukemia** • **Myelodysplasia** • Solid organ, predominantly the more common primary carcinomas
Gastrointestinal disorders
• **Inflammatory bowel disease: Crohn disease, ulcerative colitis**
Drugs
• Antibiotics (e.g. minocycline, trimethoprim-sulfamethoxazole), antihypertensives (e.g. furosemide, hydralazine), antineoplastics (e.g. ipilimumab, pembrolizumab, FLT3 inhibitors, vemurafenib), colony stimulating factors (e.g. G-CSF), contraceptives, immunosuppressant medications (e.g. azathioprine), NSAIDs, retinoids (e.g. all-*trans*-retinoic acid), bortezomid, lenalidomide
Autoimmune disorders
• Autoimmune connective tissue diseases (e.g. systemic lupus erythematosus [SLE]‡, rheumatoid arthritis, dermatomyositis, relapsing polychondritis, Sjögren syndrome), autoimmune thyroid disease • Sarcoidosis • Behçet disease
*Includes acute and chronic myelogenous leukemia, chronic lymphocytic leukemia, non-Hodgkin disease, Hodgkin's disease, myeloma, other myeloproliferative disorders. ‡Some authors use the terms non-bullous neutrophilic LE or neutrophilic dermatosis in conjunction with LE rather than Sweet syndrome associated with SLE.

CLASSIFICATION

A. CLASSIC IDIOPATHIC :
   A. INFECTION :URT → strept , git →yersinia 1-3 weeks after infection .
   B. IBD :CHRONS ,ulcerative colitis
   C. Pregnancy
B. Drug : G –CSF , AB , ANTI EPILEPTIC , OCB
C. MALIGNANCY :

2% no femal predominance

Leukemia mainly

Dissemenated recurrent, ulceration

C/P

- COSTITUTIONAL :
  - Fever →40-60 %
  - URT INFECTION
- Mucocutaneus :
  - Tender erythematous oedamatous plaque
  - Mammilated surface , pseudovesicles
  - Central yellow→ targetoid lesion
  - Head , neck , upper extremities

Fig. 26.3 Sweet syndrome. A The periocular lesion demonstrates how some lesions can mimic cellulitis (pseudocellulitis). This patient also had neutrophilic esophageal ulcerations and subsequently developed colon cancer. **B** The plaques can have a pseudomammillated appearance due to the

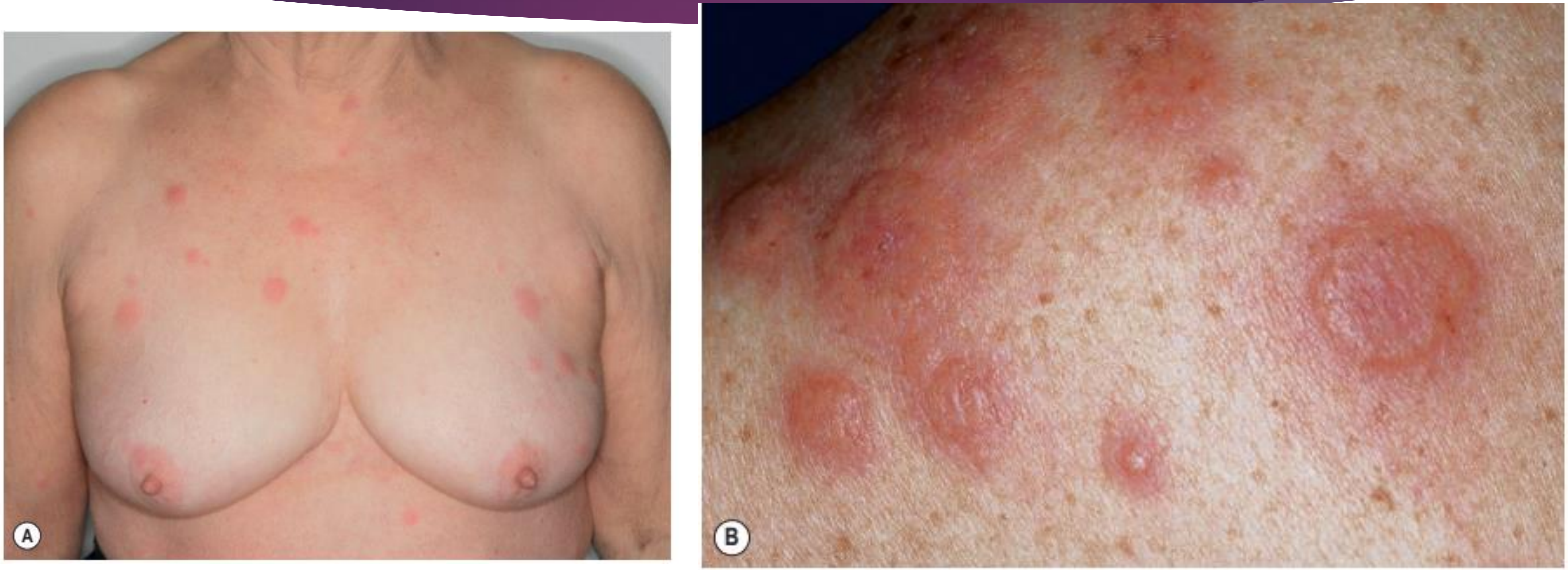


Fig. 26.2 Sweet syndrome. A Scattered edematous pink papules and plaques on the chest. B The edema can be quite marked as seen in these lesions on the

- Oral lesion :
  - Uncommon →increase in hematological
  - Pseudopustule →ulcer →aphthus
- Systemic :
  - arthralgia , myalgia , conjunctivitis .

SYSTEMIC MANIFESTATIONS OF SWEET SYNDROME
Common (≥50%)
Fever Leukocytosis
Less common (20–50%)
Arthralgias Arthritis: asymmetric, non-erosive, sterile, favors knees and wrists Myalgias Ocular involvement: conjunctivitis, episcleritis, limbal nodules, iridocyclitis
Uncommon
Neutrophilic alveolitis: cough, dyspnea and pleurisy; radiographic findings include interstitial infiltrates, nodules, pleural effusions Multifocal sterile osteomyelitis, a subtype of SAPHO (see Table 26.18) Renal involvement (e.g. mesangial glomerulonephritis): hematuria, proteinuria, renal insufficiency, acute renal failure
Unusual/rare
Acute myositis Hepatitis, pancreatitis, ileitis, colitis Aseptic meningitis, encephalitis, bilateral sensorineural hearing loss Other: aortitis, oral aphthae, pharyngitis, pharyngeal edema

Investigation :

- CBC :leukocytosis with neutrophilia
- Increase ESR , CRP
- H.P :
  - Neutrophilic infiltrate
  - Leukocytoclasia
  - Marked papillary dermal odema
  - Subepidermal vesicles
  - Endothelial swelling
  - VD
  - NO FIBRINOID NECROSIS . NO VASCULITIS

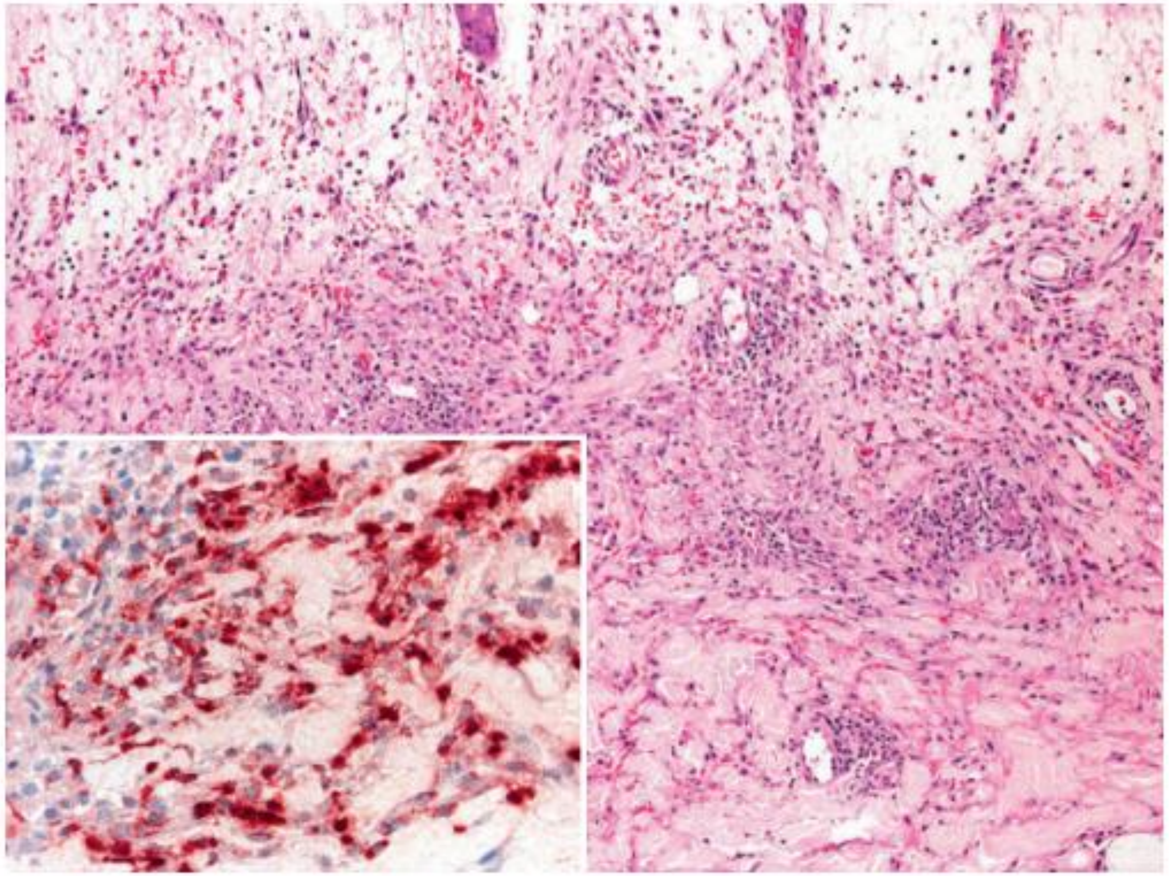


Fig. 26.6 Histiocytoid Sweet syndrome – histologic features. There is edema in the papillary dermis as well as a dermal infiltrate composed of histiocytoid cells admixed with neutrophils, a few eosinophils, and hemorrhage. The histiocytoid cells represent immature myeloid cells and therefore have stained positively for myeloperoxidase (inset). *Courtesy, Lorenzo Cerroni, MD.*

DIAGNOSTIC cRITIERIA

CRITERIA FOR DIAGNOSIS OF SWEET SYNDROME
Major criteria
1. Abrupt onset of typical cutaneous lesions 2. Histopathology consistent with Sweet syndrome
Minor criteria
1. Preceded by one of the associated infections or vaccinations; accompanied by one of the associated malignancies or inflammatory disorders; associated with drug exposure or pregnancy 2. Presence of fever and constitutional signs and symptoms 3. Leukocytosis 4. Excellent response to systemic corticosteroids

Fig. 26.7 Neutrophilic dermatosis of the dorsal hands. A Clinically and histologically, there is overlap with Sweet syndrome and bullous pyoderma gangrenosum. **B** More extensive disease is often confused with an infectious

Pyoderma gangrenos

- Non infectious neutrophilic dermatosis
- associated with systemic disease .

Fig. 26.10 Pyoderma gangrenosum (PG) – clinical presentations. A,B Typical (classic) form in which the ulcer has an undermined and overhanging violet–gray edge as well as a surrounding violaceous border. **C** Grouped sterile pustular nodules. **D** Central ulceration surrounded by inflammatory papules and pustules. **E** Deeper ulcer with vegetative and pustular base. **F** Centrally, there is healing with cribriform (sievelike) scarring.

TREATMENT

- No role for antibiotic
- Topical steroid , tarulimus
- Systemic :
  - Systemic steroid ,5 -1 mg/kg/day
    - 4-6 wks
  - K iodide →900 mg / day
  - Dapson 100 -200 mg
  - Colchecin

Aetiopathogensis

- idiopathic in 25–50%
- immunologic abnormality

Clinical variant

1. Classic ulceration :

   Tender papule→pustule →plaque →necrosis→ulceration

   Violaceus undermined edge with erythamatus rim

   - Extend centrifugaly →atrophic crepriform scar

   - +ve pathergy test
   - Solitary or multiple →lower extremities , peritibial

Pustular
Sterile pustules →limp
→IBD

Vesiculobullous
He bullae →dorsal hand
→hematological disease

Superficial granulomatous

Trunk →trauma ,surgry
→not painfull
No systemic

Fig. 26.8 Bullous pyoderma gangrenosum. This patient had ulcerative colitis.

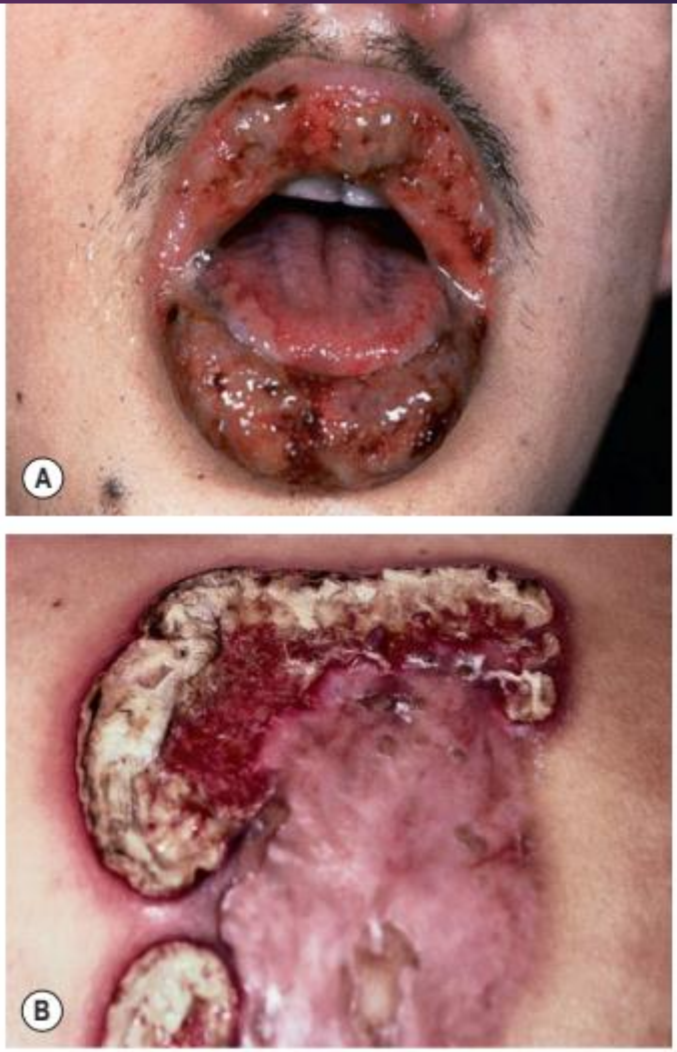


Fig. 26.9 Variants of pyoderma gangrenosum.
A Pyostomatitis vegetan in a patient with ulcerative colitis.
B Vegetative form following trauma to the skin.

CLINICAL VARIANTS OF PYODERMA GANGRENOSUM

Vesiculobullous (also referred to as atypical or bullous PG)

- Lesions favor the face and upper extremities, especially the dorsal hands
- Clinical appearance overlaps with the superficial bullous variant of Sweet syndrome (Fig. 26.8)
- Occurs most commonly in the setting of acute myelogenous leukemia, myelodysplasia, and myeloproliferative disorders such as chronic myelogenous leukemia

Pustular

- Multiple, small, sterile pustules
- Lesions usually regress without scarring, but can evolve into classic PG
- Most commonly observed in patients with inflammatory bowel disease
- Similar eruption may be seen in patients with Behçet disease or bowel-associated dermatosis–arthritis syndrome

Superficial granulomatous

- Localized, superficial vegetative or ulcerative lesion[52], which favors the trunk and usually follows trauma (e.g. surgery); may have verrucous border
- Histologically, a superficial granulomatous response with a less intense neutrophilic infiltrate
- Responds to less aggressive anti-inflammatory therapy
- Controversy as to whether it is a variant of PG, a separate disorder, or related to granulomatosis with polyangiitis (Wegener granulomatosis)

Pyoderma/pyostomatitis vegetans

- Chronic, vegetative pyoderma of the labial and buccal mucosa (Fig. 26.9A)
- May be associated with vegetative or ulcerative cutaneous PG (Fig. 26.9B)
- Large verrucous plaques that may be studded with pustules
- Seen in patients with inflammatory bowel disease

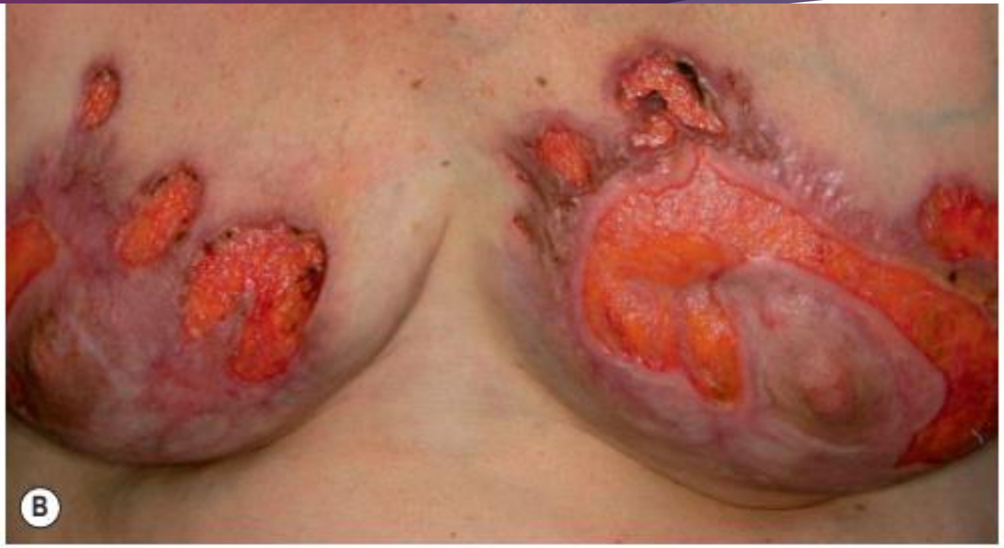
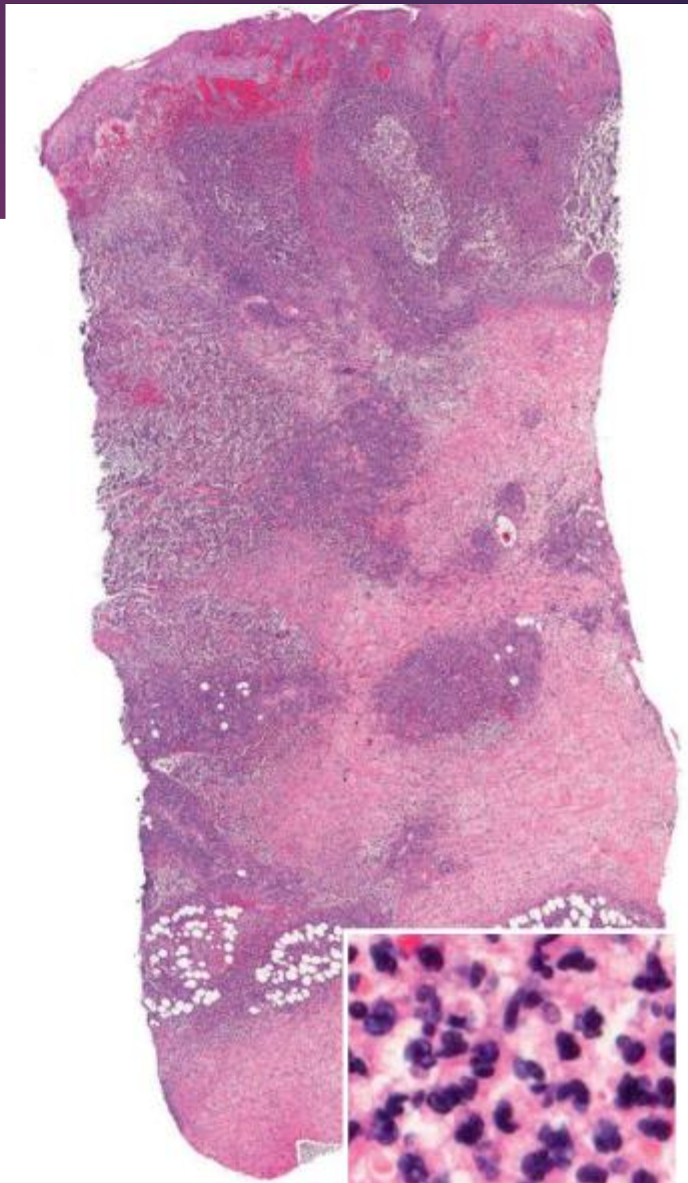


Fig. 26.11 Peristomal and postsurgical pyoderma gangrenosum (PG).
A Several ulcers surround an ileostomy in a patient who had a total proctocolectomy performed because of refractory chronic ulcerative colitis.
B Multiple ulcerations of the breasts following breast reduction. Because the original diagnosis postoperatively was soft tissue infection, multiple debridements had been performed and systemic antibiotics administered.

Association

- IBD
- ARTHRITIS
- HEMATOLOGICAL
- IGA monoclonal gammopathy



gangrenosum. In expanding untreated lesions, a diffuse infiltrate of neutrophils (inset) is present at the edge of an ulceration. *Courtesy, Lorenzo Cerroni, MD.*

- H/P
  - Non specific , non diagnostic
  - Central necrosis , ulceration
  - Neutrophilic infeltirat

- DD :
  - LEG ULCER

DIFFERENTIAL DIAGNOSIS OF PYODERMA GANGRENOSUM
Early inflammatory non-ulcerative stage (papules, pustules, plaques or nodules)
• Follicular infections (folliculitis, furuncle, carbuncle of bacterial, fungal or viral origin) • Cellulitis or soft tissue infection (bacterial, mycobacterial or fungal origin) • Insect bite reaction • Cutaneous T- and B-cell lymphomas • Halogenoderma (iododerma or bromoderma) • Panniculitides (inflammatory, infectious, metabolic, neoplastic) • Cutaneous polyarteritis nodosa • Sweet syndrome (see Table 26.5 for additional entities) • Behçet disease • Bowel-associated dermatosis–arthritis syndrome
Later ulcerative or vegetative stage
• Infections – streptococcal synergistic gangrene, ecthyma gangrenosum, atypical and typical mycobacterial infections, cutaneous lesions of the deep mycoses (e.g. blastomycosis, coccidioidomycosis, paracoccidioidomycosis, chromomycosis), botryomycosis, gummatous treponemal ulcers • Parasitic infections – leishmaniasis, amebiasis, schistosomiasis • Vascular diseases – ulcerations due to venous hypertension, arterial insufficiency, non-septic emboli, hemoglobinopathies, or thrombosis (secondary to hypercoagulability; see Ch. 105) • Vasculitis – cutaneous polyarteritis nodosa, microscopic polyangiitis, granulomatous vasculitides (granulomatosis with polyangiitis [Wegener granulomatosis], Churg–Strauss syndrome, temporal arteritis),

Diagnostic criteria

Major criteria
Rapid painfull ulcer
Exclusion of other cause

Minor criteria
+ve pathergy test
Crebriform scar
Association
H.P →neutrophilic
Rapid response to steroid

PROPOSED DIAGNOSTIC CRITERIA FOR CLASSIC ULCERATIVE PYODERMA GANGRENOSUM
Major criteria
1. Rapid[a] progression of a painful[b], necrolytic cutaneous ulcer[c] with an irregular, violaceous and undermined border 2. Other causes of cutaneous ulceration have been excluded[d]
Minor criteria
1. History suggestive of pathergy[e] or clinical finding of cribriform scarring 2. Systemic diseases associated with pyoderma gangrenosum[f] 3. Histopathologic findings (sterile dermal neutrophilia, ± mixed inflammation, ± lymphocytic vasculitis) 4. Treatment response (rapid response to systemic corticosteroids)[g]

Treatment

- No depridment →high koebnerization

1. General measure :
    1. Compression
    2. Elevation
    3. Hyperbaric o2
    4. Dressing
    5. Grafting , flab

- Mild:
- Superpotent steroid
- Tarulimus
- Ab
- Colchicine

- Sever
- Steroid 60 mg
- Cyclosporin 2,5 -5 mg
- Thalidoamid

NEUTROPHILIC DERMATOSES ASSOCIATED WITH HEMATOLOGIC MALIGNANCIES

	Sweet syndrome	Atypical (bullous) pyoderma gangrenosum	Neutrophilic eccrine hidradenitis
1. Clinical manifestations			
a. Cutaneous	Painful plaques and nodules with occasional blisters and pustules within lesions	Hemorrhagic bullae and superficial ulcerations	Erythematous edematous plaques and papules which may be painful, infrequent pustules
	Favors face, upper extremities and trunk	Favors face and upper extremities (especially dorsal hands)	Favors face, extremities, trunk
	Can recur (30–50% of patients)	Can recur	Resolves spontaneously over ~2 weeks without scarring Can recur
b. Systemic	Fever, arthralgia or arthritis Multiorgan involvement (Table 26.3) Neutrophilia Elevated ESR	Systemic features usually absent	Fever, neutropenia*
2. Histopathology	Papillary edema with dermal neutrophilic infiltrate Histiocytoid variant – histiocyte-like immature myeloid cells Lymphocytic variant – in the setting of myelodysplasia	Subepidermal hemorrhagic bullae Pustules with neutrophilic dermal infiltrate	Neutrophilic† infiltrate about and within eccrine apparatus, necrosis of eccrine epithelium, and eccrine squamous syringometaplasia
3. Associations	Infections, hematologic malignancies > solid tumors, IBD, autoimmune diseases, pregnancy, G-CSF, all-*trans*-retinoic acid (see Table 26.2)	Hematologic disorders, especially acute myelogenous leukemia and myelodysplasia, IBD, G-CSF	Leukemia with or without chemotherapy; post-chemotherapy for lymphoma and solid tumors; G-CSF; rarely infections
4. Primary treatment (in addition to treatment of underlying disease)	Systemic corticosteroids (0.5–1 mg/kg/day for 4–6 weeks), potassium iodide (900 mg/day), dapsone (100–200 mg/day), colchicine (1.5 mg/day) (see Table 26.11)	Systemic corticosteroids (see Table 26.11)	None (spontaneous resolution)

Eosinophilic dermatosis

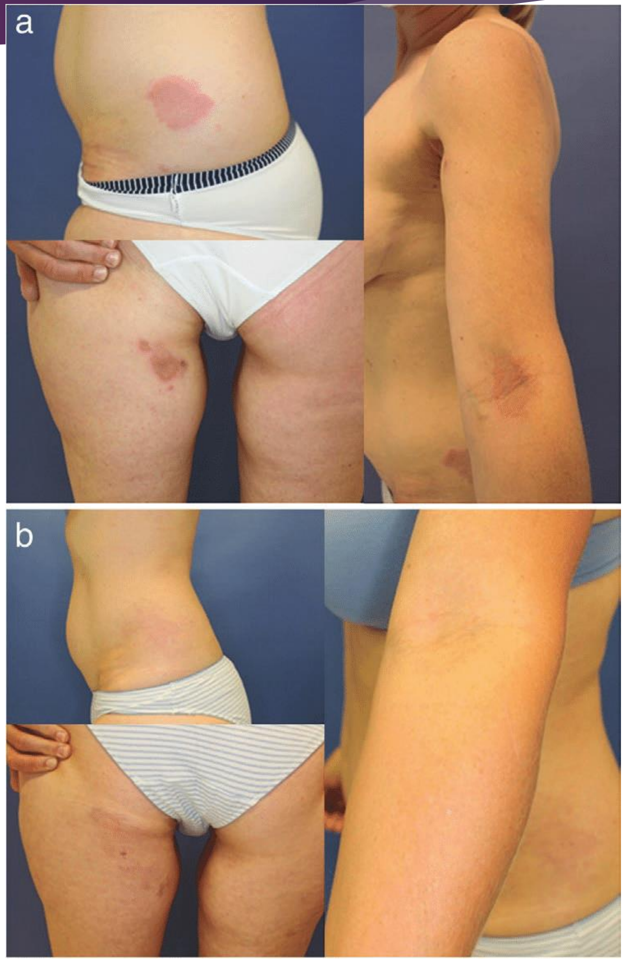
- Drug eruption
- Autoimmune bullous
- Arthropod bite
- Wells syndrome
- Parasitc infection
- Allergic disorder
- Vasculitis
- Hyper eosinophilic syndrome

Wells syndrome eosinophilic cellulitis

- Cutaneous disease with unkown etiology →
- Allergic hypersensitivity reaction →with eosinophilis
- Indurated plaque →like cellulitis →flam figure
- Rare , equal sex , all age

Triggers

Insect bite
Drug
Infection
Myloproleferative disorder



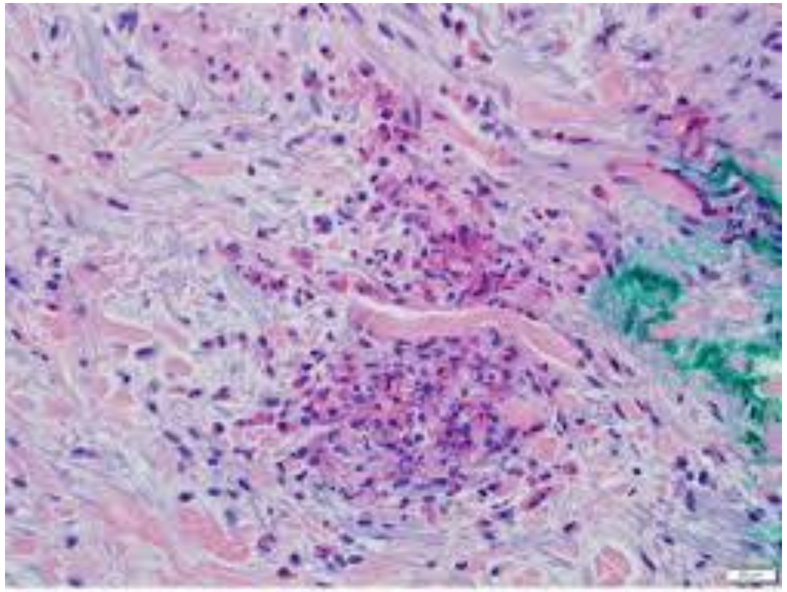
c/p

- Prodroma :fever , malais
- Bright red oedamatus nodule , plaque →
- Like urticaria , cellulitis
- →fade →brown , gray indurated
- Extremities

DD:
Cellulitis
Erythipelas
sarcoidosis

Figure 1. Annular granuloma–like variant (case 4).

- Histeopathology
- Eosinophilic dermal infilterat
- **Flam figure** :collage fiber coated with eosinophilic granules



Nodular collections of eosinophils with some degranulation on collagen fibers as could be seen in Wells syndrome

- Tt
- low dose steroid →dramatic response

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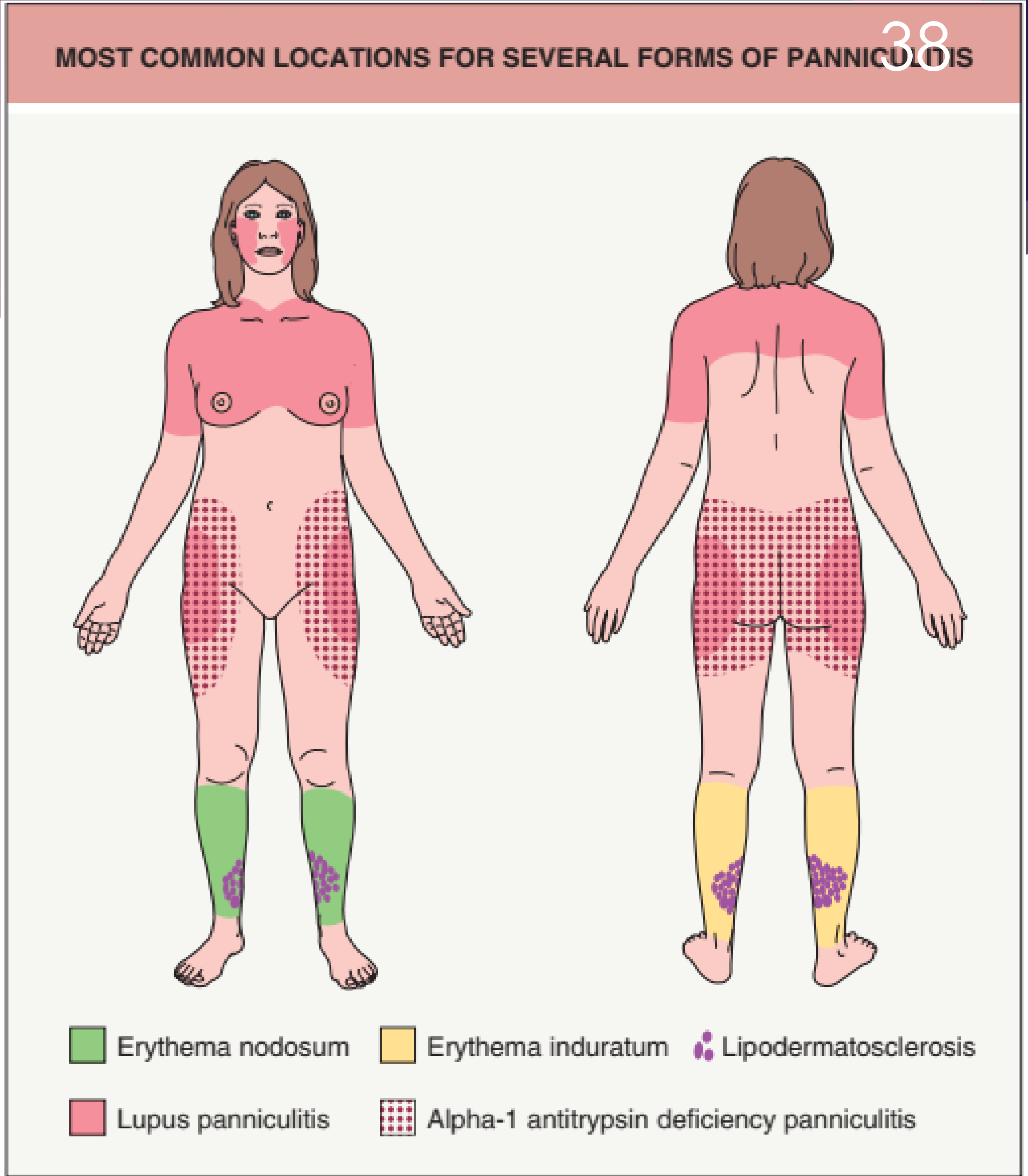
Panniculitis

CLASSIFICATION OF THE PANNICULITIDES
Predominantly septal panniculitis
• Erythema nodosum   Erythema nodosum migrans (subacute nodular migratory panniculitis) • Panniculitis of morphea/scleroderma • Alpha-1 antitrypsin deficiency panniculitis*
Lobular and mixed septal–lobular panniculitis

• With vasculitis involving septal veins or arteries, lobular venules or small veins
  Erythema induratum (nodular vasculitis)
• With necrosis as an early finding
  Pancreatic panniculitis
• With needle-shaped clefts within lipocytes
  Sclerema neonatorum
  Subcutaneous fat necrosis of the newborn
  Post-steroid panniculitis
• Associated with autoimmune connective tissue disease
  Lupus erythematosus panniculitis (lupus profundus)
  Panniculitis of dermatomyositis
• Lipodystrophic panniculitis (see Ch. 101)
  Lipoatrophy
  Lipohypertrophy
• Traumatic panniculitis, including factitial panniculitis
  Cold panniculitis (popsicle panniculitis, Haxthausen disease)
  Sclerosing lipogranuloma (including grease gun granuloma)
  Panniculitis due to other injectable substances
  Panniculitis due to blunt trauma
• Lipodermatosclerosis
• Infection-induced panniculitis
• Malignancy-related panniculitis-like infiltrates
  Subcutaneous panniculitis-like T-cell lymphoma (see Ch. 120)
  Other lymphomas with involvement of subcutaneous tissues

AN APPROACH TO THE HISTOPATHOLOGIC DIAGNOSIS OF PANNICULITIS	
Histopathologic aspects to consider	Conclusions
Look for the "center of gravity" of the infiltrate	Recommended originally by Pinkus, the purpose is to determine whether or not the inflammatory process is centered in the subcutis (favoring a primary panniculitis), in the dermis, or in the fascia (in which case the panniculitis may be a secondary manifestation of a deeper inflammatory process)
Determine if the panniculitis is predominantly septal, predominantly lobular, or mixed	Generally, a predominantly septal panniculitis narrows the diagnostic considerations (see Table 100.1). If the panniculitis is lobular or mixed, look for other distinguishing features
Determine if there is vasculitis involving a medium-sized vessel, in the face of a lobular or mixed panniculitis	This is characteristic of **erythema induratum (nodular vasculitis);** caveat: deeper levels are often required in order to demonstrate the involvement of vessels
Look for fat necrosis with saponification and "calcium soap" formation	This is a feature of **pancreatic panniculitis**
Look for needle-shaped clefts within lipocytes	Their presence favors a diagnosis of **sclerema neonatorum, subcutaneous fat necrosis of the newborn**, or **post-steroid panniculitis**
Determine if the panniculitis has a lymphoplasmacytic predominance	This tends to be a feature of **panniculitis due to connective tissue disease**, including lupus erythematosus; beware of lymphocytic predominance in cases of subcutaneous panniculitis-like T-cell lymphoma (see Table 100.9)
Check for a "central nidus" of inflammation or evidence of a needlestick injury. Look for vacuolated spaces or foreign material	These are clues to **traumatic panniculitis including factitial panniculitis**. Polarization microscopy can be helpful
Determine if lobular panniculitis has a predominance of neutrophils	**Infection-induced panniculitis, neutrophilic panniculitis** associated with inflammatory bowel disease and rheumatoid arthritis, **subcutaneous Sweet syndrome, traumatic panniculitis**; less so in alpha-1 antitrypsin deficiency panniculitis and pancreatic panniculitis
Look for membranocystic changes in the subcutis	This change is characteristic of (though not pathognomonic for) **lipodermatosclerosis;** it can be observed in any panniculitis with prominent degenerative changes


MOST COMMON LOCATIONS FOR SEVERAL FORMS OF PANNICULITIS
Erythema nodosum
Erythema induratum
Lipodermatosclerosis
Lupus panniculitis
Alpha-1 antitrypsin deficiency panniculitis

PANNICULITIDES WITH NEEDLE-SHAPED CLEFTS IN THE SUBCUTIS						
Condition	Patient characteristics	Onset	Cutaneous findings	Associated systemic findings	Histopathology	Possible precipitating factors
Sclerema neonatorum	Severely ill premature neonates	First week of life	Widespread, diffuse hardening, sparing only the genitalia, palms and soles; cool, waxy, rigid and board-like skin with purple mottled discoloration	Respiratory difficulties, congestive heart failure, intestinal obstruction, diarrhea; death, usually from septicemia, in three-quarters of patients	Needle-shaped clefts in lipocytes; septal thickening; inflammation sparse to absent (few neutrophils, eosinophils, macrophages or multinucleated giant cells may be seen)	Hypothermia, perinatal asphyxia, defective complement activity, dehydration
Subcutaneous fat necrosis of the newborn	Full-term neonates	First 2–3 weeks of life	Circumscribed, erythematous, indurated subcutaneous nodules favoring the cheeks, shoulders, back, buttocks, thighs; may become fluctuant	Hypercalcemia (onset may be delayed for several months), thrombocytopenia, hypertriglyceridemia; severe hypercalcemia may be associated with fever, eosinophilia, and persistent nephrocalcinosis	Lobular panniculitis with neutrophils, lymphocytes, macrophages; needle-shaped clefts in a radial array within lipocytes and giant cells; sometimes foci of calcification, hemorrhage	Hypothermia (including exposure to cooling blankets), hypoglycemia (e.g. due to gestational diabetes), perinatal hypoxemia (e.g. due to meconium aspiration, placenta previa, umbilical cord prolapse, pre-eclampsia)
Post-steroid panniculitis	Children (ages 1–14 years)	1–40 days* after cessation of high-dose systemic corticosteroids	Firm red plaques on the cheeks, arms, trunk; small and discrete or large and confluent lesions; pruritic, tender or asymptomatic	Underlying conditions treated with systemic corticosteroids have included leukemia, cerebral edema, nephrotic syndrome, secretory diarrhea, and acute rheumatic fever as well as	Lobular panniculitis with lymphocytes, macrophages, multinucleated giant cells; needle-shaped clefts in lipocytes and giant cells	Follows rapid withdrawal of systemic corticosteroids

Erythema nodosum

- Delayed hyoer sensetivty reaction to variety of antigenic stimulus
- Triggers
  1. Idiopathic
  2. Infection :
     1. Strept URT INFECTION → MOST common
     2. TB , yarsinia , salmonella
     3. Viral :hbv , urt
  3. Drug :ocp , salfonamide , penicillin , iodid , TNF inhibitor
  4. Sarcoidosis
  5. IBD
  6. Pregnancy
  7. Malignancy

CAUSES OF ERYTHEMA NODOSUM

Incidence	Cause	Comments
Most common	Idiopathic	Still the largest single category, accounting for a third to a half of cases
	Streptococcal infections, especially of the upper respiratory tract	The largest single infectious cause
	Other infections: Viral upper respiratory tract infections Bacterial gastroenteritis – *Yersinia* > *Salmonella, Campylobacter*	Overall, infection may account for a third or more of cases
	Coccidioidomycosis	Erythema nodosum is associated with a lower incidence of disseminated disease
	Drugs*	Especially estrogens and oral contraceptive pills; also sulfonamides, penicillin, bromides, iodides; occasionally, TNF inhibitors, BRAF inhibitors^
	Sarcoidosis**	10–20% of cases in some series
	Inflammatory bowel disease	Crohn disease has a stronger association with erythema nodosum than does ulcerative colitis
Uncommon	Infections: *Brucella melitensis* *Chlamydophila*† *pneumonia* *Chlamydia trachomatis* *Mycoplasma pneumoniae* *Mycobacterium tuberculosis* *Histoplasma capsulatum* Hepatitis B virus‡	
	Neutrophilic dermatoses: Behçet disease Sweet syndrome	"Erythema nodosum" in Behçet disease more closely resembles erythema induratum (nodular vasculitis)
	Acne fulminans, including isotretinoin-associated	
	Pregnancy	
Rare	Pernicious anemia	
	Diverticulitis	
	Infections: *Neisseria gonorrhoeae*, *N. meningitidis* *Escherichia coli*, *Bartonella henselae*, *Bordetella pertussis* *Treponema pallidum* Dermatophytes (kerion), *Blastomyces dermatitidis* HIV Giardiasis, amoebiasis (abscesses)	*Erythema nodosum leprosum* is a different disease that is characterized by a cutaneous small vessel vasculitis

- Tender erythematous sc nodules chin of tibia (anti aspect) →bilateral
- No ulceration
- Fever , malaise, arthritis
- Heal with bruis like
- No scring

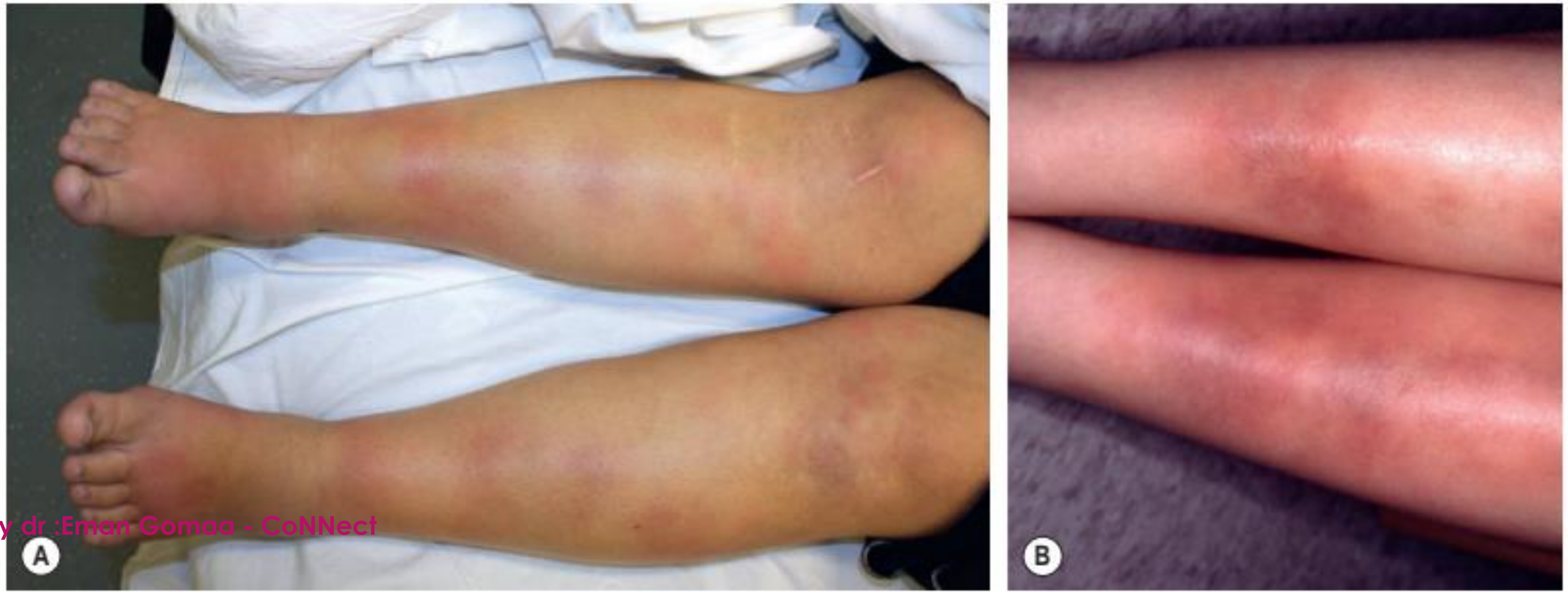


Fig. 100.2 Erythema nodosum – clinical appearance. A Erythematous, tender nodules bilaterally on the shins and dorsal feet; the patient is pregnant. **B** The nodules and plaques may develop a bruise-like appearance. *A, Courtesy, Ian Odell, MD, PhD; B, Courtesy, Kalman Watsky, MD.*

H.P

- DEEP SKIN BIOPSY
- Septal panniculitis without vasculitis
- Miescher microgranules
- 3 stages :
  - Lymphoneutrophilic →histiocytic granuloma →fibrosis

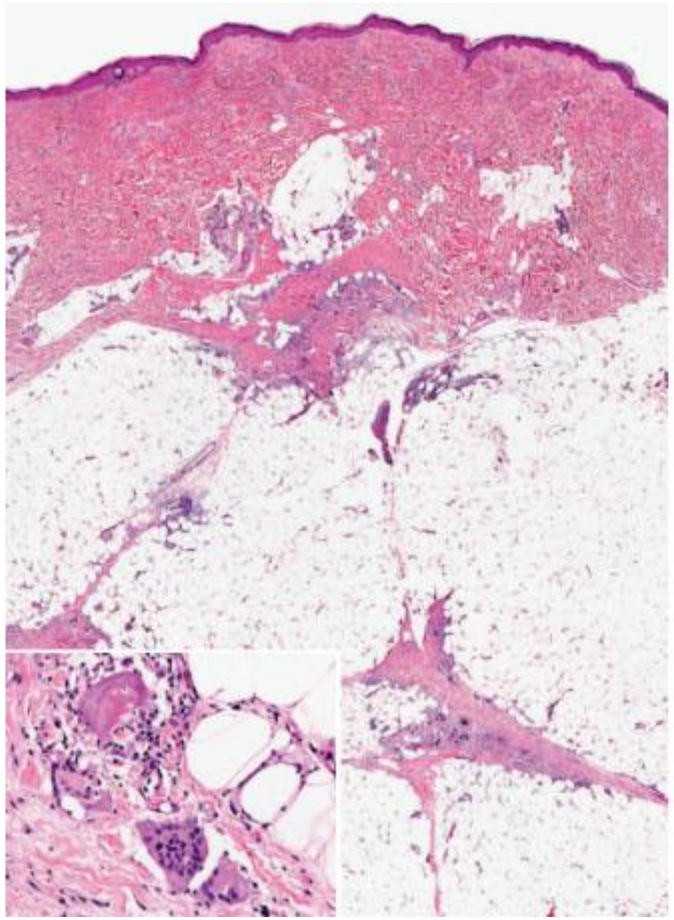


Fig. 100.3 Erythema nodosum – histopathologic features. A predominantly septal panniculitis with Miescher microgranulomas within the septa. Note the multinucleated giant cells (inset). *Courtesy, Lorenzo Cerroni, MD.*

- DD :
  - Sweet
  - EN . migrans
  - Erythema induratum of bazune
  - PAN
  - EN LEPROSUM

Mangment

Evaluation

- History : cough , dyspnea
- Asot
- CXR
- Quantiferon tb

Treatment

- Tt underlying cause
- Brd rest
- Antiinflammatory |
  - NSAID
  - K Iodid
  - Colchecin

TREATMENT RECOMMENDATIONS FOR ERYTHEMA NODOSUM	Rx
In all patients	Discontinue possible causative medications Diagnose and treat underlying cause Bed rest and leg elevation Compression
First-line	Nonsteroidal anti-inflammatory medications (3)* Salicylates Potassium iodide (2) (see Table 100.6)
Second-line**	Colchicine (3)# Infliximab (3)## Hydroxychloroquine (3)^ Adalimumab (3)^ Etanercept Mycophenolate mofetil (3)
Third-line**	Systemic corticosteroids (3) Thalidomide (3)^^ Cyclosporine (3) Dapsone (3)

K- IODIDE

- 1000mg /ml dropper
- Dose :
  - 300 mg tid
  - Children :150 mg tid
- Mechanisim
  - decrease cell mediated immunity
  - Decrease neutrophilis

- Indication
  - Sweet syndrome
  - EN
  - Sporotricosis
- S.E
  - Csvv,
  - nausea
  - Urticaria
  - Angiooedema
  - Hypothyroidisim ,

USE OF POTASSIUM IODIDE (KI)
Saturated solution of potassium iodide (SSKI)
• 1000 mg/ml • Droppers are supplied with calibrations for: 0.3 ml (300 mg)* 0.6 ml (600 mg) • In adults and older children, common dose = 300 mg TID po with starting dose = 150–300 mg TID • In infants and young children, common dose = 150 mg TID po • SSKI should be diluted in water or juice to try to minimize the bitter aftertaste • Crystallization may occur with cold temperatures, but rewarming and shaking dissolves the crystals; discard if solution turns yellow–brown
Side effects of SSKI
• Acute – nausea, bitter eructation, excessive salivation, urticaria, angioedema, cutaneous small vessel vasculitis • Chronic – enlargement of salivary and lacrimal glands, acneiform eruption, iododerma, hypothyroidism, hyperkalemia, occasionally hyperthyroidism

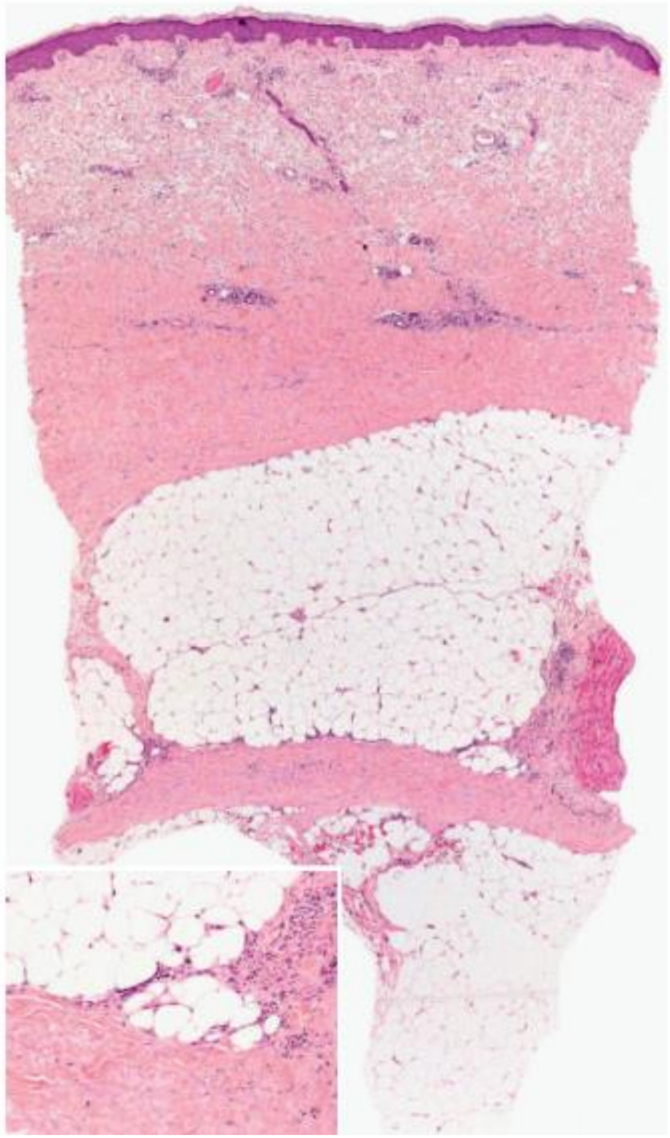


Fig. 100.4 Morphea panniculitis. Septal thickening, mucin deposition and mild lymphocytic infiltrat Lymphoplasmacytic infiltrates concentrat the dermal–subcutaneous interf (inset). *Courtesy, Lorenzo Cerroni, MD.*

Fig. 100.7 Erythema induratum – clinical appearance. Inflamed nodular lesions on the lower leg, with evidence of ulceration. *Courtesy, Kenneth E Greer, MD.*

Fig. 100.5 Alpha-1 antitrypsin deficiency panniculitis – clinical appearance. Purpuric nodules on the ankle. *Courtesy, Kenneth E Greer, MD.*

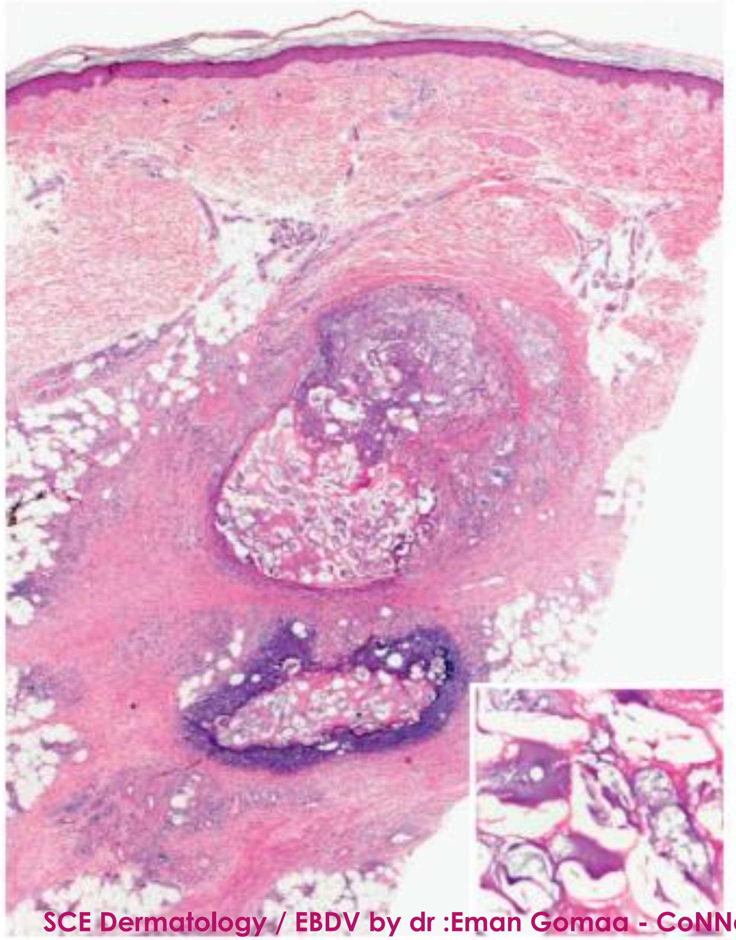


Fig. 100.10 Pancreatic panniculitis – histopathologic features. Lobular and septal panniculitis with neutrophilic inflammation, cellular necrosis, ghost cells (inset), and deposition of homogeneous basophilic material due to saponification of fat by calcium salts. *Courtesy, Lorenzo Cerroni, MD.*

Fig. 100.9 Pancreatic panniculitis – clinical appearance. Multiple red to violet–brown nodules on the arm. There is also postinflammatory desquamation.

Fig. 100.12 Subcutaneous fat necrosis of the newborn – clinical appearance. Red–violet indurated plaques, primarily on the back and shoulders.
Courtesy, Lorenzo Cerroni, MD.

3 **Lipodermatosclerosis. A** Acute phase with tender erythem… both lower extremities. **B** Chronic phase with sclerotic red–l… the lower medial leg. *B, Courtesy, Kenneth E Greer, MD.*

INFECTION-INDUCED PANNICULITIS – REPORTED CAUSATIVE AGENTS
Bacteria
Staphylococcus aureus Group A *Streptococcus* *Pseudomonas aeruginosa* *Brucella melitensis* *Nocardia asteroides* *Tropheryma whipplei* (causative agent of Whipple disease)
Mycobacteria
M. tuberculosis *M. marinum, M. avium-intracellulare, M. chelonae, M. fortuitum, M. ulcerans*
Coxiella
C. burnetii
Borrelia
B. burgdorferi
Fungi
Blastomyces dermatitidis, Histoplasma capsulatum *Microsporum* spp. (including *M. canis*)

Perforating dermatosis

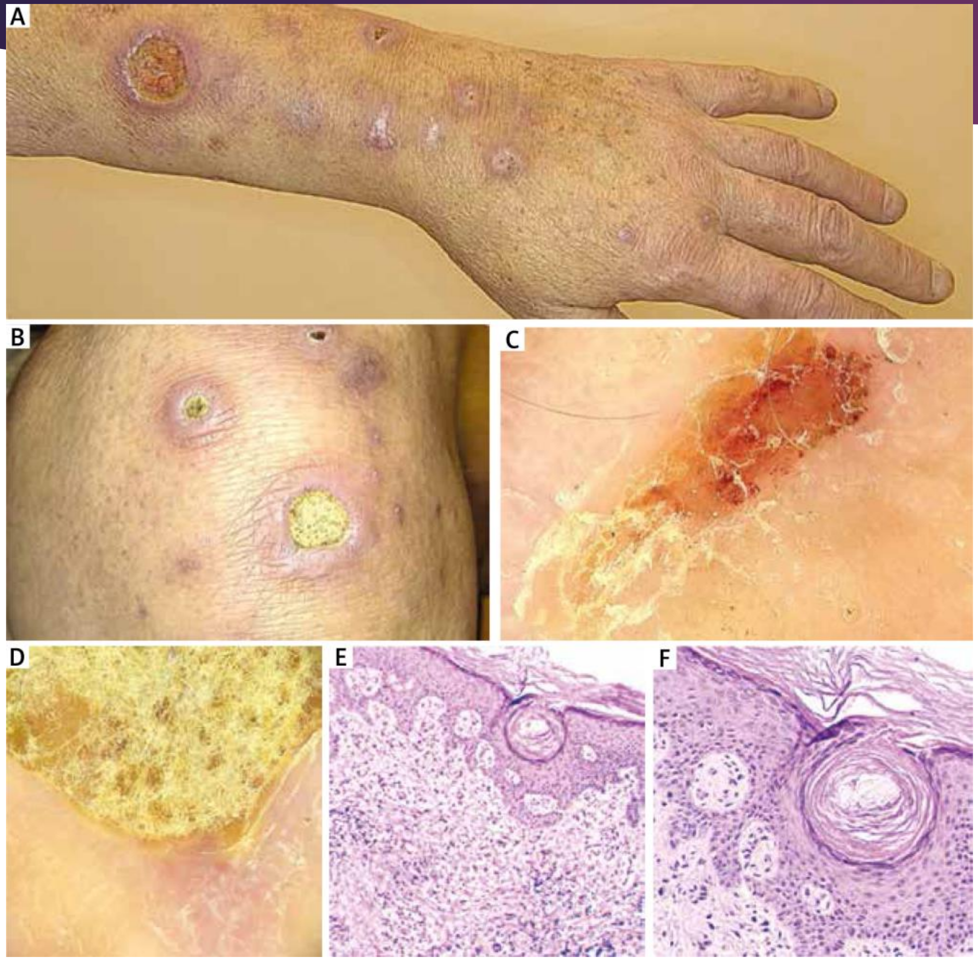
1ry perforating disease

- Kyrles disease
- Perforating folliciularis
- Reactive perforating collagenosis
- Elastosis perforans serpegenosa

2ry perforating

1. Granuloma
2. Endogenous substance
3. Endogenous F.b
4. Infectious
5. Tumor

Kyrles disease	**Perforating folliculitis**	**Reactive perforating collagenosis**	**Elastosis perfor serpegenosa**
Chronic renal failure DM	Non specific	Familial , traumatic acquired	Childhood , ad MADPOR
Papulo-nodules with central keratotic core • Bilateral , symmetrical • Leg ,extensor • Spare mucous membrane • Adult	Ordinary folliculitis with follicular rupture . Trunk , extrematies	Keratotic papules →arm ,hand Koebner →linear	Serpegenous , keratotic papul Neck Asymptomaic Pruritic , persist
Cornified plug embedded in epidermal invagination Epithelial hyperplasia Keratin , parakeratin	Compact ortho, para keratin plug Collagen , elastin	Cup shaped verticly orientedchannel with transepidermal elemenation →of collagen fiber	Epidermal elimi eosinophilic of



Perforating folliculitis

Reactive perforating collagenosis

Elastosis perforans serpegenosa


Thank you